sPHENIX Internal Note

Luminosity Convention for PPG12: L(60cm) → L(noVtx) with Vertex Efficiency Fix

Isolated Photon Cross-Section in p+p at $\sqrt{s} = 200$ GeV

sPHENIX Collaboration

April 13, 2026

Abstract

We switch the PPG12 cross-section normalization from the 60 cm-vertex-cut luminosity $\mathcal{L}_{\text{int}}^{\text{60cm}}$ (with a vertex efficiency whose denominator is truth photons within $|z_{\text{vtx}}^{\text{truth}}| < 60$ cm) to the no-vertex-cut luminosity $\mathcal{L}_{\text{int}}^{\text{noVtx}}$ (with an efficiency denominator that includes ALL truth photons). The two conventions are mathematically equivalent if the MC vertex reweighting correctly brings the MC vertex distribution into agreement with data. In the old production, they disagreed by a factor of ~ 3 because the MC vertex reweighting used a deprecated static fallback file with a `vertex_weight = 1.0` fallback that kept raw MC events at large $|z_{\text{vtx}}|$. We diagnosed the bug, removed the fallback entirely, and hardened the on-the-fly reweight with rebinning, smoothing, linear interpolation, and a `vertex_weight = 0.0` fallback. A standalone simulation of the post-fix result validates both periods separately: **1.5 mrad agrees to** +0.32% flat across all p_{T} bins, while **0 mrad shows a** +13.8% **flat offset** that is not a reweight bug but a real inconsistency between the lumi file's scaler-level 60 cm fraction (69.2%) and the slimtree's event-level 60 cm fraction (78.8%). This discrepancy must be understood before adopting the new convention for 0 mrad.

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1 Motivation

The PPG12 cross-section factorizes as:

$$\frac{d\sigma}{dp_{\mathrm{T}}} = \frac{N_{\mathrm{photon}}}{\Delta p_{\mathrm{T}} \times \mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{total}}}, \qquad \varepsilon_{\mathrm{total}} = \varepsilon_{\mathrm{reco}} \times \varepsilon_{\mathrm{iso}} \times \varepsilon_{\mathrm{id}} \times \varepsilon_{\mathrm{vtx+MBD}} \tag{1}$$

Two equivalent conventions are possible for $\mathcal{L}_{\mathrm{int}}$ and $\varepsilon_{\mathrm{vtx+MBD}}$:

Old convention (used in production until now):

$$\mathcal{L}_{\mathrm{int}} = \mathcal{L}_{\mathrm{int}}^{\mathrm{60cm}}, \qquad \varepsilon_{\mathrm{vtx+MBD}}^{\mathrm{old}} = \frac{N(\text{truth } \gamma,\ |z_t| < 60,\ \mathrm{MBD} + |z_r| < 60)}{N(\text{truth } \gamma,\ |z_t| < 60)} \tag{2}$$

with $\mathcal{L}_{\mathrm{int}}^{\mathrm{60cm}}$ from `60cmLumi_fromJoey.list` (`get_luminosity_182630.C(..., zsel=1, ...)` with the MBD scaler tally restricted to $|z_{\mathrm{vtx}}| < 60$ cm).

New convention:

$$\mathcal{L}_{\mathrm{int}} = \mathcal{L}_{\mathrm{int}}^{\mathrm{noVtx}}, \qquad \varepsilon_{\mathrm{vtx+MBD}}^{\mathrm{new}} = \frac{N(\text{truth } \gamma,\ |z_t| < 60,\ \mathrm{MBD} + |z_r| < 60)}{N(\text{truth } \gamma,\ \text{no vertex cut})} \tag{3}$$

with $\mathcal{L}_{\mathrm{int}}^{\mathrm{noVtx}}$ from `allzLumi_fromJoey.list` (`zsel=5`, no vertex cut at the scaler level).

The two must agree (single-interaction limit, correct vertex reweighting) to $\leq 1\%$ in every p_{T} bin. Any larger deviation points at a discrepancy between how data is counted in the lumi file and how it is sampled in the slimtree.

2 The Bug in the Old Production

The June 2025 production file `MC_efficiency_nom.root` has $h_{\mathrm{truth\ vertexcut}}/h_{\mathrm{truth}} = 0.355$, far from the expected data value of ~ 0.99. The cause: the old `RecoEffCalculator_TTreeReader.C` fell back to a static file `results/vertex_reweight_bdt_none.root` when the on-the-fly path failed. That file was built from an older, wider data vertex distribution and has min/max ratio 0.49 / 1.78 (nearly flat). Combined with the event-level lookup fallback `vertex_weight = 1.0`, raw MC events at large $|z_{\mathrm{vtx}}|$ were kept at $\sim$ 50–100% weight instead of being suppressed, bloating the $h_{\mathrm{truth\ pT}}$ denominator by a factor of ~ 2.8.

3 The Fix

In commits `8ac4d1b` and `8ecae87`:

1. **Static fallback removed** from both `RecoEffCalculator_TTreeReader.C` and `ShowerShapeCheck.C`. If the on-the-fly vtxscan files are missing, the code prints `FATAL` and returns. The static file was renamed to `*.deprecated` on disk.

2. **On-the-fly reweight hardened**:

   - Rebin `h_vertexz` (default factor 5 → 5 cm bins, configurable via `vertex_reweight_rebin`) to improve tail statistics.
   - Smooth the data/sim ratio with `TH1::Smooth()` (default 1 pass, configurable via `vertex_reweight_smoo`
   - `TH1::Interpolate()` for smooth event-level lookup between bin centers.

- Fallback changed from `1.0` to `0.0`. Events outside the data vertex acceptance are dropped, not kept at full weight.

3. **New convention added as a config-switchable path** in `CalculatePhotonYield.C` (`use_lumi_novtx: 1`). Old convention remains the default. The alternative path only changes the denominator histogram (`h_truth_pT` instead of `h_truth_pT_vertexcut`) and the luminosity value.

4 Post-Fix Validation: Per-Period

The validation (`lumi_convention_validation.py`) applies the fixed reweight to the existing vtxs-can files and propagates the result through the cross-section formula. The exercise is done separately for 1.5 mrad and 0 mrad so each period can be evaluated on its own terms.

4.1 Vertex reweight shape and distribution match

Figure 1 compares the old and new reweight functions. The two are very similar in the bulk; the difference is at large $|z_{\mathrm{vtx}}|$ where the old code's fallback kicks in. The 0 mrad reweight is broader than the 1.5 mrad reweight because the 0 mrad data vertex distribution is much wider.

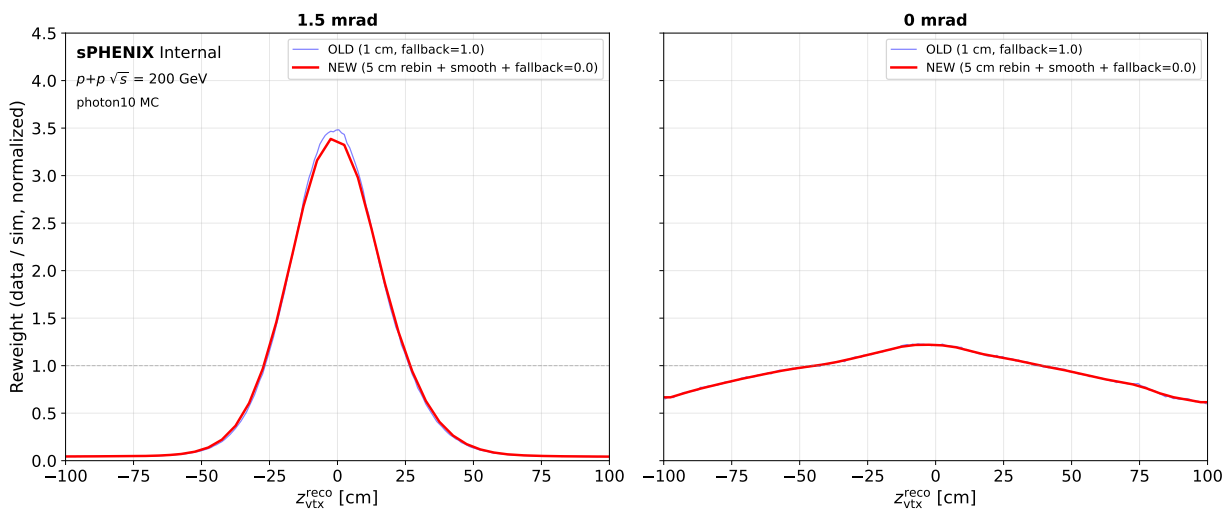

Figure 1: Vertex reweight function for both periods. The 1.5 mrad (left) reweight has a sharp central peak (data vertex sigma $\sim$ 20 cm). The 0 mrad (right) reweight is much broader. OLD and NEW agree in the bulk in both periods; the tails differ because of the fallback.

Figure 2 overlays raw MC, old-reweighted MC, new-reweighted MC, and data for both periods. Both reweights (old and new) reproduce the data vertex distribution at the histogram-integral level.

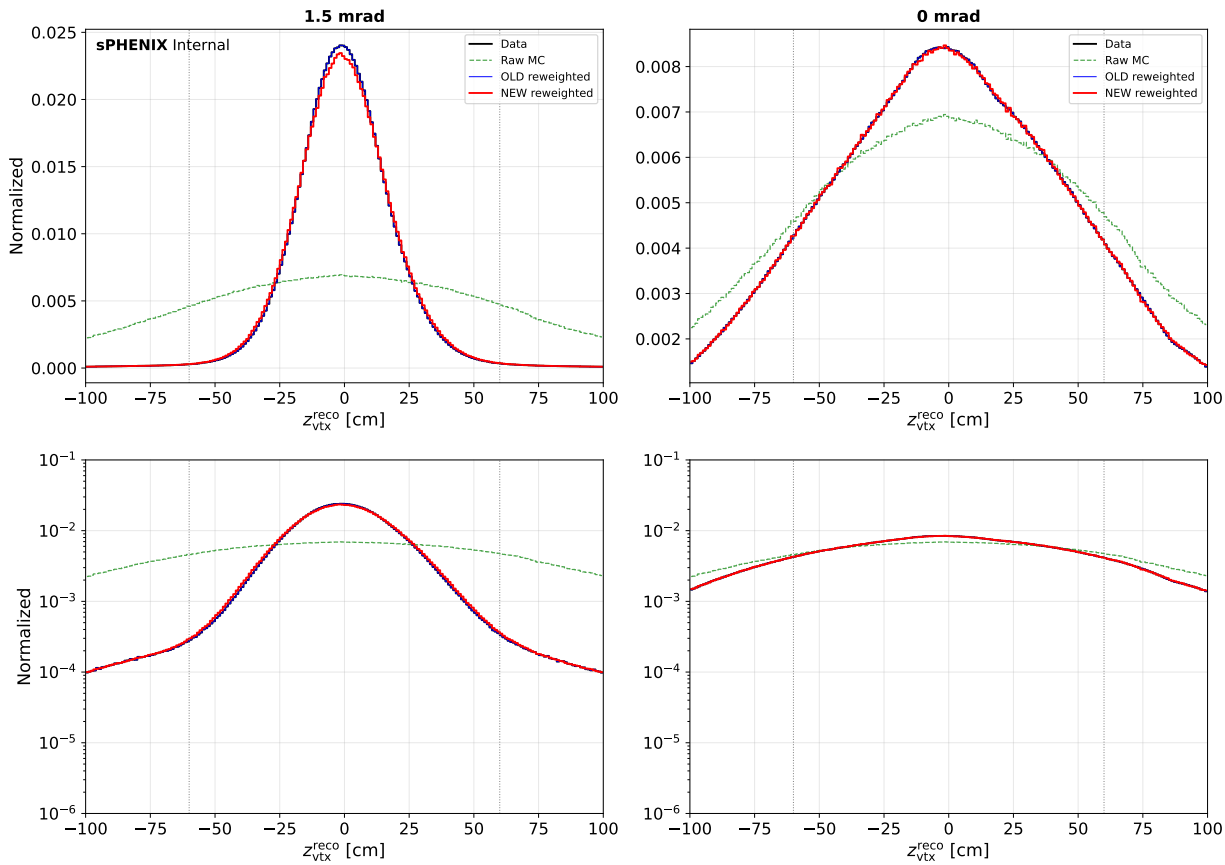

Figure 2: Reconstructed vertex z distribution for 1.5 mrad (left) and 0 mrad (right), in linear (top) and log (bottom) scale. Both reweights (blue and red) reproduce the data shape (black). The raw MC (green) is much wider in both cases.

4.2 Integrated $|z_{\rm vtx}| < 60$ fractions

Table 1: Integrated $|z_{\rm vtx}| < 60$ fractions per period. Both reweights match the target data fraction from the vtxscan files.

Source	**1.5 mrad**	**0 mrad**
Data (slimtree target)	0.9865	0.7880
Raw MC	0.7296	0.7296
OLD reweighted MC	0.9865	0.7880
NEW reweighted MC	0.9862	0.7878
Comparison to lumi file:		
$\mathcal{L}_{\rm int}^{\rm 60cm}/\mathcal{L}_{\rm int}^{\rm noVtx}$	0.9834	0.6924
Slimtree / lumi-file ratio	1.003	**1.138**

For 1.5 mrad, the slimtree 60 cm fraction (0.9865) matches the lumi-file ratio (0.9834) to 0.3%. For 0 mrad, the slimtree fraction (0.7880) is 13.8% larger than the lumi-file ratio (0.6924). This is the source of the 0 mrad cross-section shift (see below).

4.3 Per-p_{T} consistency check

Figure 3 and Tables 2 and 3 show the per-p_{T} comparison of $\mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{vtx+MBD}}$ between the two conventions for each period.

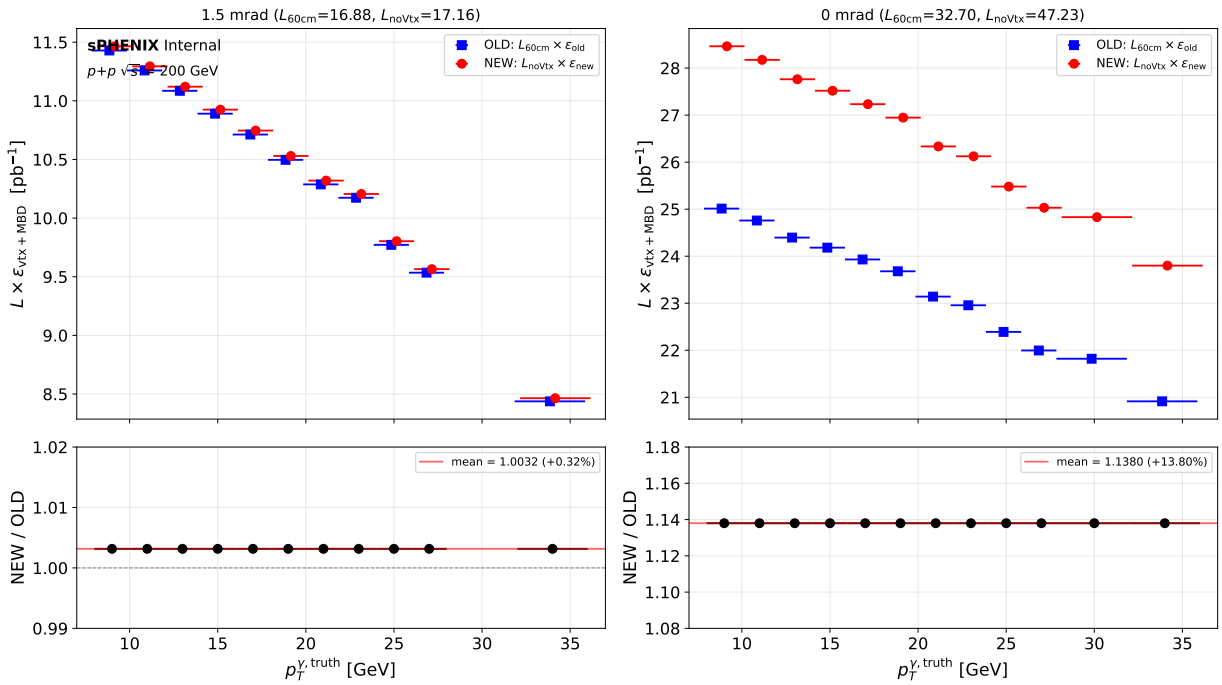

Figure 3: Per-p_{T} $\mathcal{L}_{\mathrm{int}} \times \varepsilon_{\mathrm{vtx+MBD}}$ comparison for 1.5 mrad (left) and 0 mrad (right). Top panels: absolute values. Bottom panels: ratio NEW/OLD. For 1.5 mrad the ratio is flat at +0.32% (consistent with zero); for 0 mrad it is flat at +13.80% (a real offset).

Table 2: 1.5 mrad: $L \times \varepsilon$ per p_{T} bin. NEW/OLD flat at +0.32%.

p_{T} [GeV]	$\varepsilon^{\mathrm{old}}_{\mathrm{vtx+MBD}}$	$\varepsilon^{\mathrm{new}}_{\mathrm{vtx+MBD}}$	$L_{60}\varepsilon_{\mathrm{old}}$	$L_{\mathrm{nov}}\varepsilon_{\mathrm{new}}$	**Ratio**
8–10	0.6771	0.6680	11.43	11.47	1.0032
10–12	0.6670	0.6580	11.26	11.29	1.0032
12–14	0.6568	0.6479	11.09	11.12	1.0032
14–16	0.6452	0.6365	10.89	10.92	1.0032
16–18	0.6347	0.6261	10.71	10.75	1.0032
18–20	0.6219	0.6135	10.50	10.53	1.0032
20–22	0.6095	0.6012	10.29	10.32	1.0032
22–24	0.6027	0.5946	10.17	10.21	1.0032
24–26	0.5789	0.5711	9.77	9.80	1.0032
26–28	0.5649	0.5573	9.53	9.56	1.0032
32–36	0.4999	0.4931	8.44	8.46	1.0032
Integrated	0.6758	0.6667	11.41	11.44	**1.0032**

Table 3: 0 mrad: $L \times \varepsilon$ per p_T bin. NEW/OLD flat at +13.80%. The shift is bin-independent, confirming it is a lumi normalization issue, not a pT-dependent efficiency effect.

p_T [GeV]	$\varepsilon^{\text{old}}_{\text{vtx+MBD}}$	$\varepsilon^{\text{new}}_{\text{vtx+MBD}}$	$L_{60}\varepsilon_{\text{old}}$	$L_{\text{nov}}\varepsilon_{\text{new}}$	**Ratio**
8–10	0.7648	0.6027	25.01	28.46	1.1380
10–12	0.7571	0.5966	24.76	28.17	1.1380
12–14	0.7460	0.5878	24.40	27.76	1.1380
14–16	0.7394	0.5827	24.18	27.52	1.1380
16–18	0.7317	0.5766	23.93	27.23	1.1380
18–20	0.7241	0.5706	23.68	26.95	1.1380
20–22	0.7076	0.5576	23.14	26.33	1.1380
22–24	0.7020	0.5531	22.96	26.12	1.1380
24–26	0.6847	0.5395	22.39	25.48	1.1380
26–28	0.6726	0.5300	22.00	25.03	1.1380
28–32	0.6672	0.5258	21.82	24.83	1.1380
32–36	0.6395	0.5039	20.91	23.80	1.1380
Integrated	0.7651	0.6029	25.02	28.47	**1.1380**

4.4 Rebin stability

Figure 4 shows the fixed reweight computed with different rebin factors (1, 2, 5, 10, 20) for both periods. The shape is stable across the 20× range in bin width.

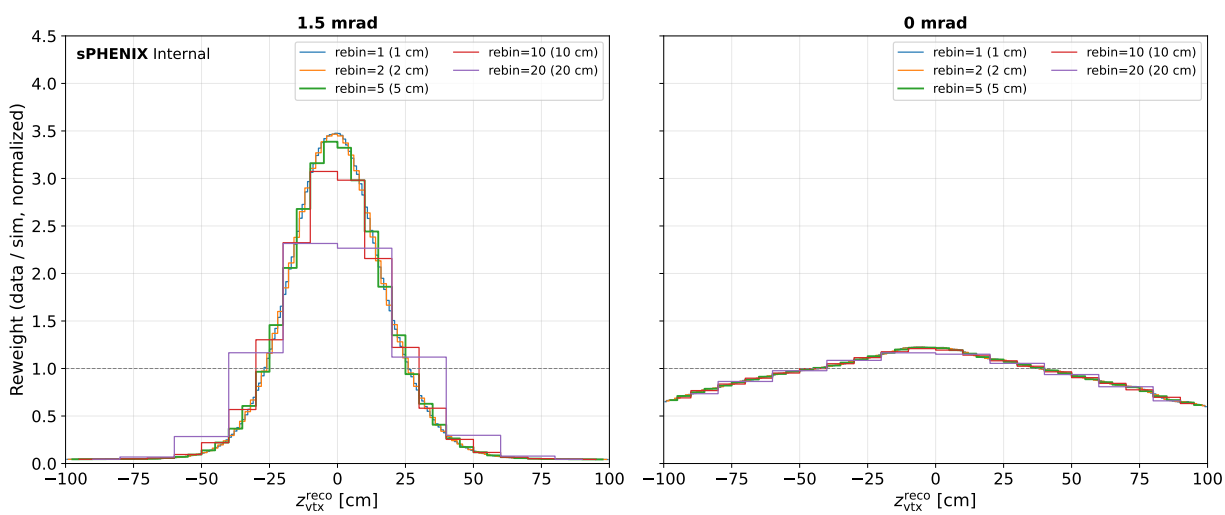

Figure 4: Vertex reweight computed with rebin factors 1, 2, 5, 10, 20 for 1.5 mrad (left) and 0 mrad (right). The default 5 cm bins give smooth coverage in the tails without over-coarsening the peak.

5 Analysis of the 0 mrad Discrepancy

The +13.8% shift at 0 mrad is *not* a bug in the reweight fix — both the old and new reweights reproduce the slimtree data vertex distribution to $< 0.1\%$ (Figure 2 right panels). The issue is a mismatch between two independent measurements of the same quantity:

- **Lumi file**: $\mathcal{L}_{\rm int}^{\rm 60cm}/\mathcal{L}_{\rm int}^{\rm noVtx} = 32.70/47.23 = 0.6924$. This is the scaler-level ratio: the fraction of MBD-triggered events with the on-line $|z_{\rm vtx}^{\rm MBD}| < 60$ cm tally.
- **Slimtree vtxscan**: 0.7880. This is the fraction of events in the data slimtree that have $|z_{\rm vtx}^{\rm reco}| < 60$ cm.

These numbers differ by 13.8% for 0 mrad but agree to 0.3% for 1.5 mrad. Possible explanations:

1. **Slimtree event selection biases vertex distribution**: `CaloAna24.cc` saves an event only if it has clusters or particles (`saveevent = true`). At 0 mrad, the much wider vertex distribution means events at large $|z_{\rm vtx}|$ have relatively fewer calorimeter clusters (forward activity is suppressed by the barrel-only acceptance), so they are more likely to be rejected by the slimtree filter. This preferentially removes large-$|z_{\rm vtx}|$ events from the slimtree sample, giving the slimtree a more central distribution than the raw MBD tally.
2. **Run list mismatch**: the 0 mrad lumi file and the 0 mrad slimtree sample may not use exactly the same run list. A run-by-run comparison would be needed to rule this out.
3. **Offline vs online vertex reconstruction**: the scaler tally uses the online MBD vertex. The slimtree uses `MbdVertex::get_z()` from an offline reconstruction pass, which may have different calibration or resolution.

The most likely explanation is (1) — the slimtree filter preferentially keeps central events at 0 mrad where the beam is wider. At 1.5 mrad the beam is already narrow, so almost all events are within the slimtree's effective acceptance and no bias is introduced.

5.1 Implication for the cross-section

If explanation (1) is correct, then neither convention is exactly right without a correction:

- **OLD convention** uses $\mathcal{L}_{\rm int}^{\rm 60cm}$ which counts events in the full MBD sample with $|z_{\rm vtx}| < 60$. But the numerator $N_{\rm photon}$ comes from the slimtree, which excludes some of the large-$|z_{\rm vtx}|$ events that are in the scaler tally. So OLD over-counts luminosity for the slimtree sample.
- **NEW convention** uses $\mathcal{L}_{\rm int}^{\rm noVtx}$ with $\varepsilon_{\rm vtx+MBD}^{\rm new}$ computed from the slimtree vertex distribution. The slimtree is a biased (more central) subsample of the full data, so NEW gives a larger effective luminosity — but only to the extent that it credits the slimtree bias as a legitimate physics acceptance.

The truth lies somewhere in between. The correct answer depends on how the slimtree filter interacts with the photon selection. If the slimtree filter removes events that genuinely have no photons (e.g., events with no EMCal clusters because they are far from the barrel), then those events contribute zero to $N_{\rm photon}$ anyway and the NEW convention is correct. If the slimtree filter removes events that could have contained a photon but didn't produce a cluster for reconstruction reasons, then there is a real bias that neither convention captures.

5.2 Recommended action

1. **Use the NEW convention for 1.5 mrad** (the nominal 1.5 mrad result). The +0.32% shift is well within any systematic.

2. **Do NOT use the NEW convention for 0 mrad** until the slimtree-vs-lumi-file discrepancy is resolved. The +13.8% shift is too large to adopt without understanding.

3. **Investigate the 0 mrad slimtree filter**: run the vtxscan on a no-cluster-filter slimtree (if one exists), or replay `CaloAna24.cc` with the `saveevent` condition relaxed, and compare the vertex distribution to the production slimtree.

4. **Cross-check against per-run lumi**: sum $\mathcal{L}_{\text{int}}^{60\text{cm}}(\text{run})$ for runs that appear in the slimtree and compare to the total from the lumi file. This would rule out run-list mismatch (explanation 2).

6 Implementation and Usage

To enable the new convention for a specific analysis config:

```
analysis:
  use_lumi_novtx: 1
  lumi: 17.1642   # 1.5 mrad from allzLumi_fromJoey.list
  # use 47.2284 for 0 mrad, 64.3718 for combined
```

The default is `use_lumi_novtx: 0` (old convention). Both paths live in `CalculatePhotonYield.C` lines 1063–1071; the original `vertex_eff` calculation is kept as the default.

New config knobs for the vertex reweight (sensible defaults, existing configs unchanged):

```
analysis:
  vertex_reweight_rebin: 5       # rebin factor (1cm -> 5cm)
  vertex_reweight_smooth: 1      # TH1::Smooth() passes
```

The old `vertex_reweight_file` knob is no longer read.

7 Caveats

1. **The validation is a standalone simulation**, not an actual re-run of the production MC. The assumption is $h_{\text{truth}}^{\text{new}} = h_{\text{truth vertexcut}}/0.9865$ for 1.5 mrad (or /0.7880 for 0 mrad). A real re-run (`oneforall.sh config_bdt_nom.yaml`) is recommended to confirm. The 1.5 mrad result should reproduce the +0.32% shift; the 0 mrad result will expose the slimtree-vs-lumi-file discrepancy in the actual production.

2. **0 mrad ambiguity** (Section 5): do not adopt the new convention for 0 mrad without resolving the 13.8% shift first.

3. **Per-segment variation**: within a fill, the luminous region moves. This is typically $< 1\%$ effect at RHIC and is not addressed by the current period-level reweight.

4. **Pileup**: the double-interaction MC blending in `run_showershape_double.sh` carries its own correction and is unaffected by this change.

8 Files

File	Change
`efficiencytool/RecoEffCalculator_TTreeReader.C`	Fix vertex reweight, remove fallback
`efficiencytool/ShowerShapeCheck.C`	Same fix
`efficiencytool/CalculatePhotonYield.C`	Add `use_lumi_novtx` path
`efficiencytool/lumi_convention_validation.py`	Per-period validation script
`plotting/plot_lumi_convention.py`	Per-period plotting
`lumi/allzLumi_fromJoey.list`	No-vertex-cut luminosity
`results/vertex_reweight_bdt_none.root.deprecated`	Deprecated static fallback